

Ibrahim Khebour

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Education

Colorado State University <i>PhD in Computer Science</i>	Aug 2022 – Present Fort Collins, CO
Tunisia Polytechnic School <i>Multidisciplinary Engineering Degree</i>	Sep 2019 – Jun 2022 Tunis, Tunisia
Preparatory Institute for Engineering Studies of El Manar <i>Ranked 36th/2000 (Top 2%)</i>	Sep 2017 – Jun 2019 Tunis, Tunisia

Research Experience

Graduate Research Assistant <i>Colorado State University, SIGNAL Lab</i>	Aug 2022 – Present Fort Collins, CO
– Architecting multimodal ML systems combining speech, vision, and behavioral signals for common ground tracking. – Designing real-time inference pipelines for low-latency deployment in live environments. – Contributing to NSF iSAT and DARPA FACT projects.	
Graduation Research Intern <i>Colorado State University</i>	Feb 2022 – Jun 2022 Fort Collins, CO
– Developed multilingual loanword detection models across 10+ languages. – Applied data augmentation for low-resource NLP tasks.	
Data Science Intern <i>Datagram</i>	Jun 2021 – Aug 2021 Paris, France
– Built semantic similarity systems using Transformer models (BERT, RoBERTa, USE).	

Publications

- Khebour, Ibrahim, Jung, Changsoo, Fitzgerald, Jack, Jamil, Huma, Krishnaswamy, Nikhil. **Feature Contributions to Multimodal Interpretation of Meaning**. Proceedings of the International Conference on Human-Computer Interaction, 2025.
- Jamil, Huma, Khebour, Ibrahim, Lai, Kenneth, Pustejovsky, James, Krishnaswamy, Nikhil. **A Graph Autoencoder Approach for Gesture Classification with Gesture AMR**. Proceedings of the International Conference on Computational Semantics, 2025.
- Zhu, Yifan, Lai, Kenneth, Khebour, Ibrahim, Verhagen, Marc, Krishnaswamy, Nikhil, Pustejovsky, James. **Multimodal Situational Awareness: Neuro-Symbolic AI for Real-Time HCI in the Classroom**. Proceedings of the International Conference on Human-Computer Interaction, 2025.
- Bradford, Mariah, Khebour, Ibrahim, VanderHoeven, Hannah, Venkatesha, Videep, Blanchard, Nathaniel, Krishnaswamy, Nikhil. **Tracking Individual Beliefs in Co-situated Groups Using Multimodal Input**. Proceedings of the International Conference on Human-Computer Interaction, 2025.

- Venkatesha, Videep, Nath, Abhijnan, **Khebour, Ibrahim**, Chelle, Avyakta, Bradford, Mariah, Tu, Jingxuan, VanderHoeven, Hannah, Bhalla, Brady, Youngren, Austin, Fitzgerald, Jack, et al. **Propositional Extraction from Collaborative Naturalistic Dialogues**. Journal of Educational Data Mining, 2025.
- VanderHoeven, Hannah, Bhalla, Brady, **Khebour, Ibrahim**, Youngren, Austin, Venkatesha, Videep, Bradford, Mariah, Fitzgerald, Jack, Mabrey, Carlos, Tu, Jingxuan, Zhu, Yifan, et al. **TRACE: Real-Time Multimodal Common Ground Tracking in Situated Collaborative Dialogues**. Proceedings of the North American Chapter of the Association for Computational Linguistics (System Demonstrations), 2025.
- Palmer, Derek, Zhu, Yifan, Lai, Kenneth, VanderHoeven, Hannah, Bradford, Mariah, **Khebour, Ibrahim**, Mabrey, Carlos, Fitzgerald, Jack, Krishnaswamy, Nikhil, Palmer, Martha, et al. **Speech Is Not Enough: Interpreting Nonverbal Indicators of Common Knowledge and Engagement**. Proceedings of the AAAI Conference on Artificial Intelligence, 2025.
- VanderHoeven, Hannah, Bradford, Mariah, Jung, Changsoo, **Khebour, Ibrahim**, Lai, Kenneth, Pustejovsky, James, Krishnaswamy, Nikhil, Blanchard, Nathaniel. **Multimodal Design for Interactive Collaborative Problem-Solving Support**. Proceedings of the International Conference on Human-Computer Interaction, 2024.
- Zhu, Yifan, VanderHoeven, Hannah, Lai, Kenneth, Bradford, Mariah, Tam, Christopher, **Khebour, Ibrahim**, Brutti, Richard, Krishnaswamy, Nikhil, Pustejovsky, James. **Modeling Theory of Mind in Multimodal HCI**. Proceedings of the International Conference on Human-Computer Interaction, 2024.
- Venkatesha, Videep, Nath, Abhijnan, **Khebour, Ibrahim**, Chelle, Avyakta, Bradford, Mariah, Tu, Jingxuan, Pustejovsky, James, Blanchard, Nathaniel, Krishnaswamy, Nikhil. **Propositional Extraction from Natural Speech in Small Group Collaborative Tasks**. International Conference on Educational Data Mining, 2024.
- **Khebour, Ibrahim**, Lai, Kenneth, Bradford, Mariah, Zhu, Yifan, Brutti, Richard, Tam, Christopher, Tu, Jingxuan, Ibarra, Benjamin, Blanchard, Nathaniel, Krishnaswamy, Nikhil, et al. **Common Ground Tracking in Multimodal Dialogue**. Proceedings of the Joint International Conference on Computational Linguistics, Language Resources and Evaluation, 2024.
- **Khebour, Ibrahim**, Brutti, Richard, Dey, Indrani, Dickler, Rachel, Sikes, Kelsey, Lai, Kenneth, Bradford, Mariah, Cates, Brittany, Hansen, Paige, Jung, Changsoo, et al. **When Text and Speech Are Not Enough: A Multimodal Dataset of Collaboration in a Situated Task**. Journal of Open Humanities Data, 2024.
- Terpstra, Corbyn, **Khebour, Ibrahim**, Bradford, Mariah, Wisniewski, Brett, Krishnaswamy, Nikhil, Blanchard, Nathaniel. **How Good is Automatic Segmentation as a Multimodal Discourse Annotation Aid?** Proceedings of the Joint ACL-ISO Workshop on Interoperable Semantics, 2023.
- Bradford, Mariah, **Khebour, Ibrahim**, Blanchard, Nathaniel, Krishnaswamy, Nikhil. **Automatic Detection of Collaborative States in Small Groups Using Multimodal Features**. Proceedings of the International Conference on Artificial Intelligence in Education, 2023.

- Nath, Abhijnan, Saravani, Sina Mahdipour, **Khebour, Ibrahim**, Mannan, Sheikh, Li, Zihui, Krishnaswamy, Nikhil. **A Generalized Method for Automated Multilingual Loanword Detection**. Proceedings of the International Conference on Computational Linguistics, 2022.

Specialized Skills

- **Programming:** Python (PyTorch, HuggingFace), C/C++, R, SQL, LaTeX
- **Core Areas:** Multimodal ML, NLP, Real-Time Systems
- **Mathematics:** Linear Algebra, Probability, Optimization
- **Languages:** English, French, Arabic